JSS MAHAVIDHYAPEETHA

JSS SCIENCE AND TECHNOLOGY UNIVERSITY

SRI JAYACHAMARAJENDRA COLLEGE OF ENGINEERING

- Constituent College of JSS Science and Technology University
- Approved by A.I.C.T.E
- Governed by the Grant-in-Aid Rules of Government of Karnataka
- Identified as lead Institution for World Bank Assistance under TEQIP Scheme

DEPARTMENT OF MECHANICAL ENGINEERING

SCHEME & SYLLABUS

First / Second Semester

(From the Academic Year 2024-25)

Course Title	**ELEMENTS OF MECHANICAL ENGINEERING**			**Course Type**		**Theory**			
Course Code	**24ME111/ 24ME211**	**Credits**	**03**	**Class**		**I/II**			
Course Structure	TLP	Credits	Contact Hours	Total Number of Classes /Semester		**Assessment in Weightage and Marks**			
	Theory	03	3	Theory	Practical	CIE		SEE	
	Practical					**Weightage**	40%	**Weightage**	60%
	Tutorial					**Max. Marks**	40 Marks	**Max. Marks**	60 Marks
	Total	**03**	**3**	**39**		**Min. Marks**	20 Marks	**Min. Marks**	25 Marks

Course Overview: Students are introduced to the fundamentals of Mechanical Engineering.

Pre-requisites: Nil

Course Objective:

1	To explain formation of steam, its properties and working principle of boilers.
2	To illustrate the working of heat engines and water turbines.
3	To describe power transmission and metal joining process.
4	To explain the working principles of machine tools and its operations.
5	To illustrate the working of refrigeration, air conditioning, robotics and automation.

Course Outcome:

CO#	Course Outcome	Highest Level of Cognitive Domain
CO1	Explain formation of steam, its properties and working principle of boilers.	L2
CO2	Illustrate the working of heat engines and water turbines.	L2
CO3	Describe power transmission and metal joining process.	L2
CO4	Explain the working principles of machine tools and its operations.	L2
CO5	Illustrate the working of refrigeration, air conditioning, robotics and automation.	L2

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 - Create

Course Articulation:

Course Articulation:

CO #	PROGRAM OUTCOMES												PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3													3
CO2	3	3													3
CO3	3	3											3		
CO4	3	3											3		
CO5	3	3											2		3

High – 3, Medium – 2, Low – 1

Course Content / Syllabus:

UNIT No.	Content	Hours	
		Lecture	T
1	**Steam Formation:** Steam formation and its representation on T-S diagram, Steam Properties – Specific Volume, Enthalpy, External Work of Evaporation and Internal Energy. **Boilers**: Classification, working principle of water and fire tube boilers. List and significance of Boiler Mountings and Accessories.	07	
2	**Heat Engines:** Classification, Working Principle of Two and Four Stroke Petrol and Diesel Engines. Gas turbines -Classification, Working Principle of Open Cycle and Closed Cycle Gas Turbines. **Water turbines:** Classification, Working Principle of Pelton wheel and Kaplan turbine.	08	
3	**Belt and gear Drives** – Working of Open and Crossed belt drive, Velocity Ratio, Creep and Slip.Terminology used in gears , Types of Gear Drives, Spur, Helical, Bevel, Worm & Worm Wheel and Rack &	08	

	Pinion.Simple and Compound Gear Trains. **Metal Joining Processes:** Definition of Welding, Soldering and Brazing, Working principle of Arc Welding and Gas Welding. Types of flames in gas welding.		
4	**Lathe** - Parts and Specification of Centre Lathe, Lathe Operations such as Turning, Facing, Knurling, Thread Cutting, Drilling, Taper turning by Swiveling Compound Rest Method. **Drilling Machine** - Parts of Drilling Machine. Types of drilling machine, working of Bench Drilling Machine, Drilling Machine Operations such as Drilling, Boring, Reaming, Tapping, Counter Sinking and Counter Boring.	08	
5	**Refrigeration and Air conditioning:** Refrigerants, Properties of Refrigerants. Working Principle of Vapour Compression Refrigeration and Vapour Absorption Refrigeration Systems, Working Principle of Air conditioner. **Robotics and Automation -**Fixed, Programmable and Flexible automation. Configuration of robot, advantages, limitations and applications of robots.	08	

Text Books:

1. Elements of Mechanical Engineering - Kestoor Praveen, Ramesh M R:InterlinePublishing House.

Reference Books:

1. ElementsofMechanicalEngineering-HajraChoudhury&others,MediaPromoters 2010
2. Elementsof Mechanical Engineering Sciences – Dr. K.V.A. Balaji &
3. K.Ramasastry,SanguinePublications2006.
4. TheElementsofWorkshopTechnology-VolI&II,S.K.HajraChoudhury,
5. A.K.HajraChoudhury,NirjharRoy,11thedition2001others,MediaPromotersand Publishers, Mumbai.
6. A Text Book of Elements of Mechanical Engineering – by K R Gopala Krishna , Sudheer Gopala Krishna , S C Sharma , January 2015, Subhash Publishers, Bangalore

Web/Digital resources:

1. LearnMech2.ASME3. iMechanica4.Coursera - shorturl.at/kGIOR5.edX6. Swayam.ac.in
2. Nptel.ac.in - shorturl.at/kACLW

Evaluation Scheme:

Continuous Internal Evaluation – CIE

Event	Event Type	Marks Allotted	Duration
CIE – 1	Written Test – 1	30	1 Hour
CIE – 2	Event	20	1 Hour

CIE – 3	Written Test – 2	30	1 Hour

Note:

1. The written test 1 & 2 (CIE – 1 & 3) both will be conducted for 30 marks each in 1-hour duration and the marks scored will be reduced proportionately to 15 marks each.
2. The Event (CIE – 2) will be conducted for 20 marks and the marks scored will be reduced proportionately to 10 marks.
3. The Event (CIE – 2) will be skill-based assessment such as Seminars / Technical talks / Case study / hands-on activity / Mini projects / Sci-tech activity / Data analysis.
4. A student must score on an average of 50% i.e., 20 marks out of 40 from all the events (CIE - 1, 2, 3) to gain the eligibility to appear for SEE.

Semester End Examination – SEE

Event	Event Type	Marks Allotted	Duration
SEE	Written Examination	100	3 Hours

Note:

1. The SEE will be conducted for 100 marks and the marks scored will be proportionately reduced to 60 marks. A student must score a minimum of 25 marks out of 60 in the SEE to pass.
2. SEE Question paper will be set for 100 marks and will have two parts (Part-A & Part-B). Questions under Part-A are compulsory and questions under Part-B will have internal choices.